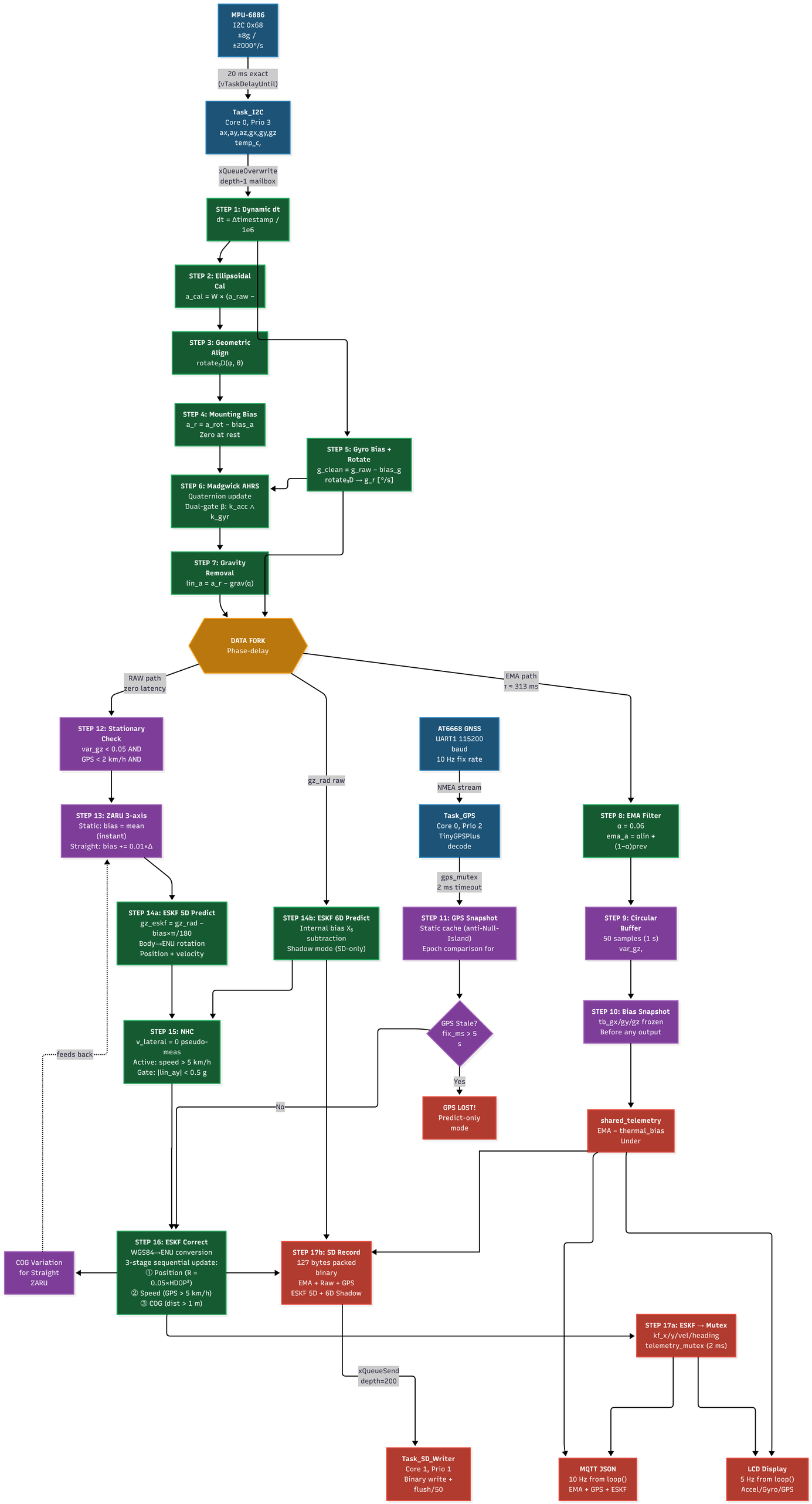
MPU-6886
I2C 0x68
±8g /
±2000°/s
20 ms exact
(vTaskDelayUntil)
Task_I2C
Core 0, Prio 3
ax,ay,az,gx,gy,gz
temp_c,
xQueueOverwrite
depth-1 mailbox
STEP 1: Dynamic dt
dt = Δtimestamp /
1e6
STEP 2: Ellipsoidal
Cal
a_cal = W × (a_raw −
STEP 3: Geometric
Align
rotate3D(φ, θ)
STEP 4: Mounting Bias
a_r = a_rot − bias_a
Zero at rest
STEP 5: Gyro Bias +
Rotate
g_clean = g_raw − bias_g
rotate3D → g_r [°/s]
STEP 6: Madgwick AHRS
Quaternion update
Dual-gate β: k_acc ∧
k_gyr
STEP 7: Gravity
Removal
lin_a = a_r − grav(q)
DATA FORK
Phase-delay
RAW path
zero latency
EMA path
τ ≈ 313 ms
STEP 12: Stationary
Check
var_gz < 0.05 AND
GPS < 2 km/h AND
gz_rad raw
AT6668 GNSS
UART1 115200
baud
10 Hz fix rate
NMEA stream
STEP 13: ZARU 3-axis
Static: bias = mean
(instant)
Straight: bias += 0.01×Δ
Task_GPS
Core 0, Prio 2
TinyGPSPlus
decode
STEP 8: EMA Filter
α = 0.06
ema_a = αlin +
(1−α)prev
gps_mutex
2 ms timeout
STEP 14a: ESKF 5D Predict
gz_eskf = gz_rad −
bias×π/180
Body→ENU rotation
Position + velocity
STEP 14b: ESKF 6D Predict
Internal bias X5
subtraction
Shadow mode (SD-only)
STEP 11: GPS Snapshot
Static cache (anti-Null-
Island)
Epoch comparison for
STEP 9: Circular
Buffer
50 samples (1 s)
var_gz,
STEP 10: Bias Snapshot
tb_gx/gy/gz frozen
Before any output
STEP 15: NHC
v_lateral = 0 pseudo-
meas
Active: speed > 5 km/h
Gate: |lin_ay| < 0.5 g
feeds back
GPS Stale?
fix_ms > 5
s
Yes
No
GPS LOST!
Predict-only
mode
shared_telemetry
EMA − thermal_bias
Under
COG Variation
for Straight
ZARU
STEP 16: ESKF Correct
WGS84→ENU conversion
3-stage sequential update:
① Position (R =
0.05×HDOP²)
② Speed (GPS > 5 km/h)
③ COG (dist > 1 m)
STEP 17b: SD Record
127 bytes packed
binary
EMA + Raw + GPS
ESKF 5D + 6D Shadow
STEP 17a: ESKF → Mutex
kf_x/y/vel/heading
telemetry_mutex (2 ms)
xQueueSend
depth=200
Task_SD_Writer
Core 1, Prio 1
Binary write +
flush/50
MQTT JSON
10 Hz from loop()
EMA + GPS + ESKF
LCD Display
5 Hz from loop()
Accel/Gyro/GPS